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Common Protocols

Internet protocols are standardized rules and guidelines defined in RFCs that specify how devices on a network should communicate with each other. They ensure that devices on a network can exchange information consistently and reliably, regardless of the hardware and software used. For devices to communicate on a network, they need to be connected through a communication channel, such as a wired or wireless connection. The devices then exchange information using a set of standardized protocols that define the format and structure of the data being transmitted. The two main types of connections used on networks are Transmission Control Protocol (`TCP`) and User Datagram Protocol (`UDP`).

We need to deal with and know the different and most used protocols. As we have already learned, these protocols are the basis of all communication between our devices and computers in the networks. We have compiled below many of these protocols that we will be dealing with throughout the modules. The better we understand them, the more effectively we can work with them.

Transmission Control Protocol

`TCP` is a `connection-oriented` protocol that establishes a virtual connection between two devices before transmitting data by using a Three-Way-Handshake. This connection is maintained until the data transfer is complete, and the devices can continue to send data back and forth as long as the connection is active.

For example, When we enter a URL into our web browser, the browser sends an HTTP request to the server hosting the website using `TCP`. The server responds by sending the HTML code for the website back to the browser using `TCP`. The browser then uses this code to render the website on our screen. This process relies on a `TCP` connection being established between the browser and the web server and maintained until the data transfer is complete. As a result, `TCP` is reliable but slower than UDP because it requires additional overhead for establishing and maintaining the connection.

Protocol	**Acronym**	**Port**	**Description**
Telnet	`Telnet`	`23`	Remote login service
Secure Shell	`SSH`	`22`	Secure remote login service
Simple Network Management Protocol	`SNMP`	`161-162`	Manage network devices
Hyper Text Transfer Protocol	`HTTP`	`80`	Used to transfer webpages

Protocol	Acronym	Port	Description
Hyper Text Transfer Protocol Secure	HTTPS	443	Used to transfer secure webpages
Domain Name System	DNS	53	Lookup domain names
File Transfer Protocol	FTP	20-21	Used to transfer files
Trivial File Transfer Protocol	TFTP	69	Used to transfer files
Network Time Protocol	NTP	123	Synchronize computer clocks
Simple Mail Transfer Protocol	SMTP	25	Used for email transfer
Post Office Protocol	POP3	110	Used to retrieve emails
Internet Message Access Protocol	IMAP	143	Used to access emails
Server Message Block	SMB	445	Used to transfer files
Network File System	NFS	111 , 2049	Used to mount remote systems
Bootstrap Protocol	BOOTP	67 , 68	Used to bootstrap computers

Protocol	Acronym	Port	Description
Kerberos	Kerberos	88	Used for authentication and authorization
Lightweight Directory Access Protocol	LDAP	389	Used for directory services
Remote Authentication Dial-In User Service	RADIUS	1812, 1813	Used for authentication and authorization
Dynamic Host Configuration Protocol	DHCP	67, 68	Used to configure IP addresses
Remote Desktop Protocol	RDP	3389	Used for remote desktop access
Network News Transfer Protocol	NNTP	119	Used to access newsgroups
Remote Procedure Call	RPC	135, 137-139	Used to call remote procedures
Identification Protocol	Ident	113	Used to identify user processes
Internet Control Message Protocol	ICMP	0-255	Used to troubleshoot network issues

Protocol	Acronym	Port	Description
Internet Group Management Protocol	IGMP	0-255	Used for multicasting
Oracle DB (Default/Alternative) Listener	oracle-tns	1521 / 1526	The Oracle database default/alternative listener is a service that runs on the database host and receives requests from Oracle clients.
Ingres Lock	ingreslock	1524	Ingres database is commonly used for large commercial applications and as a backdoor that can execute commands remotely via RPC.
Squid Web Proxy	http-proxy	3128	Squid web proxy is a caching and forwarding HTTP web proxy used to speed up a web server by caching repeated requests.
Secure Copy Protocol	SCP	22	Securely copy files between systems
Session Initiation Protocol	SIP	5060	Used for VoIP sessions
Simple Object Access Protocol	SOAP	80 , 443	Used for web services
Secure Socket Layer	SSL	443	Securely transfer files

Protocol	Acronym	Port	Description
TCP Wrappers	TCPW	113	Used for access control
Internet Security Association and Key Management Protocol	ISAKMP	500	Used for VPN connections
Microsoft SQL Server	ms-sql-s	1433	Used for client connections to the Microsoft SQL Server.
Kerberized Internet Negotiation of Keys	KINK	892	Used for authentication and authorization
Open Shortest Path First	OSPF	89	Used for routing
Point-to-Point Tunneling Protocol	PPTP	1723	Is used to create VPNs
Remote Execution	REXEC	512	This protocol is used to execute commands on remote computers and send the output of commands back to the local computer.
Remote Login	RLOGIN	513	This protocol starts an interactive shell session on a remote computer.
X Window System	X11	6000	It is a computer software system and network protocol that provides a

Protocol	Acronym	Port	Description
			graphical user interface (GUI) for networked computers.
Relational Database Management System	`DB2`	`50000`	RDBMS is designed to store, retrieve and manage data in a structured format for enterprise applications such as financial systems, customer relationship management (CRM) systems.

User Datagram Protocol

On the other hand, `UDP` is a `connectionless` protocol, which means it does not establish a virtual connection before transmitting data. Instead, it sends the data packets to the destination without checking to see if they were received.

For example, when we stream or watch a video on a platform like YouTube, the video data is transmitted to our device using `UDP`. This is because the video can tolerate some data loss, and the transmission speed is more important than the reliability. If a few packets of video data are lost along the way, it will not significantly impact the overall quality of the video. This makes `UDP` faster than TCP but less reliable because there is no guarantee that the packets will reach their destination.

Protocol	Acronym	Port	Description
Domain Name System	DNS	53	It is a protocol to resolve domain names to IP addresses.
Trivial File Transfer Protocol	TFTP	69	It is used to transfer files between systems.
Network Time Protocol	NTP	123	It synchronizes computer clocks in a network.
Simple Network Management Protocol	SNMP	161	It monitors and manages network devices remotely.
Routing Information Protocol	RIP	520	It is used to exchange routing information between routers.
Internet Key Exchange	IKE	500	Internet Key Exchange
Bootstrap Protocol	BOOTP	68	It is used to bootstrap hosts in a network.
Dynamic Host Configuration Protocol	DHCP	67	It is used to assign IP addresses to devices in a network dynamically.

Protocol	Acronym	Port	Description
Telnet	TELNET	23	It is a text-based remote access communication protocol.
MySQL	MySQL	3306	It is an open-source database management system.
Terminal Server	TS	3389	It is a remote access protocol used for Microsoft Windows Terminal Services by default.
NetBIOS Name	netbios-ns	137	It is used in Windows operating systems to resolve NetBIOS names to IP addresses on a LAN.
Microsoft SQL Server	ms-sql-m	1434	Used for the Microsoft SQL Server Browser service.
Universal Plug and Play	UPnP	1900	It is a protocol for devices to discover each other on the network and communicate.
PostgreSQL	PGSQL	5432	It is an object-relational database management system.
Virtual Network Computing	VNC	5900	It is a graphical desktop sharing system.
X Window System	X11	6000-6063	It is a computer software system and network protocol that provides GUI on Unix-like systems.

Protocol	Acronym	Port	Description
Syslog	SYSLOG	514	It is a standard protocol to collect and store log messages on a computer system.
Internet Relay Chat	IRC	194	It is a real-time Internet text messaging (chat) or synchronous communication protocol.
OpenPGP	OpenPGP	11371	It is a protocol for encrypting and signing data and communications.
Internet Protocol Security	IPsec	500	IPsec is also a protocol that provides secure, encrypted communication. It is commonly used in VPNs to create a secure tunnel between two devices.
Internet Key Exchange	IKE	11371	It is a protocol for encrypting and signing data and communications.
X Display Manager Control Protocol	XDMCP	177	XDMCP is a network protocol that allows a user to remotely log in to a computer running the X11.

ICMP

Internet Control Message Protocol (`ICMP`) is a protocol used by devices to communicate with each other on the Internet for various purposes, including error reporting and status information. It sends requests and messages between devices, which can be used to report errors or provide status information.

ICMP Requests

A request is a message sent by one device to another to request information or perform a specific action. An example of a request in ICMP is the `ping` request, which tests the connectivity between two devices. When one device sends a ping request to another, the second device responds with a `ping reply` message.

ICMP Messages

A message in ICMP can be either a request or a reply. In addition to ping requests and responses, ICMP supports other types of messages, such as error messages, `destination unreachable`, and `time exceeded` messages. These messages are used to communicate various types of information and errors between devices on the network.

For example, if a device tries to send a packet to another device and the packet cannot be delivered, the device can use ICMP to send an error message back to the sender. ICMP has two different versions:

- `ICMPv4`: For IPv4 only
- `ICMPv6`: For IPv6 only

ICMPv4 is the original version of `ICMP`, developed for use with IPv4. It is still widely used and is the most common version of ICMP. On the other hand, ICMPv6 was developed for IPv6. It includes additional functionality and is designed to address some of the limitations of ICMPv4.

Request Type	Description
`Echo Request`	This message tests whether a device is reachable on the network. When a device sends an echo request, it expects to receive an echo reply message. For example, the tools `tracert` (Windows) or `traceroute` (Linux) always send ICMP echo requests.
`Timestamp Request`	This message determines the time on a remote device.
`Address Mask Request`	This message is used to request the subnet mask of a device.

Message Type	Description
`Echo reply`	This message is sent in response to an echo request message.
`Destination unreachable`	This message is sent when a device cannot deliver a packet to its destination.
`Redirect`	A router sends this message to inform a device that it should send its packets to a different router.
`time exceeded`	This message is sent when a packet has taken too long to reach its destination.

Message Type	Description
`Parameter problem`	`This message is sent when there is a problem with a packet's header.`
`Source quench`	`This message is sent when a device receives packets too quickly and cannot keep up. It is used to slow down the flow of packets.`

Another crucial part of ICMP for us is the Time-To-Live (`TTL`) field in the ICMP packet header that limits the packet's lifetime as it travels through the network. It prevents packets from circulating indefinitely on the network in the event of routing loops. Each time a packet passes through a router, the router decrements the `TTL value by 1`. When the TTL value reaches `0`, the router discards the packet and sends an ICMP `Time Exceeded` message back to the sender.

We can also use `TTL` to determine the number of hops a packet has taken and the approximate distance to the destination. For example, if a packet has a `TTL` of 10 and takes 5 hops to reach its destination, it can be inferred that the destination is approximately 5 hops away. For example, if we see a ping with the `TTL` value of `122`, it could mean that we are dealing with a Windows system (`TTL 128` by default) that is 6 hops away.

However, it is also possible to guess the operating system based on the default `TTL` value used by the device. Each operating system typically has a default `TTL` value when sending packets. This value is set in the packet's header and is decremented by 1 each time the packet passes through a router. Therefore, examining a device's default `TTL` value makes it possible to infer which operating system the device is using. For example: Windows systems (`2000/XP/2003/Vista/10`) typically have a default `TTL` value of 128, while macOS and Linux systems typically have a default `TTL` value of 64 and Solaris' default `TTL` value of 255. However, it is important to note that the user can change these values, so they should be independent of a definitive way to determine a device's operating system.

VoIP

Voice over Internet Protocol (`VoIP`) is a method of transmitting voice and multimedia communications. For example, it allows us to make phone calls using a broadband internet connection instead of a traditional phone line, like Skype, Whatsapp, Google Hangouts, Slack, Zoom, and others.

The most common VoIP ports are `TCP/5060` and `TCP/5061`, which are used for the Session Initiation Protocol (SIP). However, the port `TCP/1720` may also be used by some VoIP systems for the H.323 protocol, a set of standards for multimedia communication over packet-based networks. Still, SIP is more widely used than H.323 in VoIP systems.

Nevertheless, SIP is a signaling protocol for initiating, maintaining, modifying, and terminating real-time sessions involving video, voice, messaging, and other communications applications and services between two or more endpoints on the Internet. Therefore, it uses requests and methods between the endpoints. The most common SIP requests and methods are:

Method	Description
`INVITE`	`Initiates a session or invites another endpoint to participate.`

Method	Description
`ACK`	Confirms the receipt of an INVITE request.
`BYE`	Terminate a session.
`CANCEL`	Cancels a pending INVITE request.
`REGISTER`	Registers a SIP user agent (UA) with a SIP server.
`OPTIONS`	Requests information about the capabilities of a SIP server or user agent, such as the types of media it supports.

Information Disclosure

However, SIP allows us to enumerate existing users for potential attacks. This can be done for various purposes, such as determining a user's availability, finding out information about the user's capabilities or services, or performing brute-force attacks on user accounts later on.

One of the possible ways to enumerate users is the SIP `OPTIONS` request. It is a method used to request information about the capabilities of a SIP server or user agents, such as the types of media it supports, the codecs it can decode, and other details. The `OPTIONS` request can probe a SIP server or user agent for information or test its connectivity and availability.

During our analysis, it is possible to discover a `SEPxxxx.cnf` file, where `xxxx` is a unique identifier, is a configuration file used by Cisco Unified Communications Manager, formerly known as Cisco CallManager, to define the settings and parameters for a Cisco Unified IP Phone. The file specifies the phone model, firmware version, network settings, and other details.

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